PRASANT KUMAR SAHU

Software Engineer | Full Stack Java & Generative AI

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PROFESSIONAL SUMMARY

Software Engineer with **3.5+ years** building enterprise applications with **Java, Spring Boot, Microservices, React.js and Next.js**, and hands-on experience bringing **Generative AI** into production workflows. Builds AI-powered features with **Python, LangChain, Ollama, MCP (Model Context Protocol), RAG and vector databases** for intelligent document processing, retrieval and automation at Mercedes-Benz R&D (via IBM). Combines strong backend engineering (distributed systems, Kafka, AWS, CI/CD) with LLM integration, prompt engineering and AI-assisted development to ship scalable, AI-enabled products.

TECHNICAL SKILLS

AI / GenAI: Generative AI, LLM Integration, LangChain, Ollama (local LLMs), MCP (Model Context Protocol), RAG, Vector Databases, Embeddings, Prompt Engineering

Languages: Java (8/17/21/25), Python, JavaScript, TypeScript, SQL

Backend: Spring Boot, Spring Cloud, Spring Security, Spring Data JPA, Reactive Spring, Microservices, REST APIs, Hibernate/JPA, Apache Kafka, Resilience4j

Frontend: React.js, Next.js, Redux Toolkit, HTML5, CSS3, Tailwind CSS, shadcn/ui, Material UI

Databases: PostgreSQL, MySQL, MongoDB, Redis, Vector Databases

Cloud & DevOps: AWS (EC2, S3, Lambda, Route 53, SNS, IAM, CloudWatch), Docker, Jenkins, AWS CodeBuild/CodePipeline, Kubernetes (basic), Linux

AI-Assisted Development: GitHub Copilot, Cursor, ChatGPT/LLM tools for code generation, test writing, code review and documentation

Practices: System Design, Distributed Systems, Design Patterns, DSA, JUnit, Mockito, Code Reviews, Agile/Scrum, Mentoring

PROFESSIONAL EXPERIENCE

Senior Software Developer / Application Developer | IBM India

Apr 2026 – Present

Client: Mercedes-Benz R&D India (MBRDI) – DeltaDMU | DPV (Document Processing Verification)

- Integrated **Generative AI** into automotive document workflows using **Python, LangChain and Ollama**, enabling intelligent document processing, retrieval, analysis and verification.
- Built **RAG pipelines with vector databases** to index engineering documents and serve context-aware answers to engineers, reducing manual document lookup.
- Integrated **MCP (Model Context Protocol)** to connect LLMs with enterprise tools and backend data for AI-driven automation.
- Developed enterprise applications with **Java 21/25, Spring Boot, React.js and Next.js** for document processing, verification and engineering workflows.
- Migrated Spring Boot services from **Java 17 to Java 21 and then Java 25 (LTS)**, adopting **virtual threads**, records, pattern matching for switch and scoped values to simplify concurrency and improve throughput.
- Enhanced a **3D Assembly Viewer** with historical records, DeltaDMU comparison and automated Delta calculations for engineering analysis.
- Built scalable **REST APIs and microservices** linking the AI services, data retrieval, historical tracking and the frontend.
- Applied system design for scalability, fault tolerance and performance; contributed to code reviews, CI/CD (Docker, Jenkins) and Agile/Scrum delivery.
- **Tech Stack:** Java 17/21/25, Spring Boot, Spring Security, Microservices, Python, LangChain, Ollama, MCP, RAG, Vector DB, React.js, Next.js, PostgreSQL, Redis, Kafka, AWS, Docker, Jenkins

Software Developer | Altruist Technologies Pvt Ltd

May 2025 – Apr 2026

Project: Glow Networks – HR Management Platform for large US clients (including Ciena)

- Designed and built scalable backend **microservices with Java 17, Spring Boot and Hibernate** for enterprise HR workflows.
- Built a **proof-of-concept LLM feature** for the HR platform, integrating an LLM API with Spring Boot services to automate a manual HR task and evaluate GenAI for production use.
- Wrote **Python scripts** to automate data processing, validation and reporting tasks, reducing repetitive manual work for the team.
- Used **AI coding assistants** (GitHub Copilot, Cursor, ChatGPT) daily to speed up development, generate unit tests and review code, while validating AI output for quality.
- Built secure REST APIs with validation, exception handling and **Spring Security** authentication/authorization; developed React.js and Next.js frontend components.
- Implemented **event-driven communication with Apache Kafka**; optimized PostgreSQL/MySQL queries, indexing and Hibernate mappings to cut API response times.

- Automated CI/CD with Jenkins and Docker on AWS; led technical discussions, conducted code reviews and mentored team members.
- **Tech Stack:** Java 17/21, Spring Boot, Spring Security, Hibernate, Microservices, Python, LLM APIs, React.js, Next.js, PostgreSQL, MySQL, Kafka, Docker, Jenkins, AWS

Associate System Engineer | Envotions Technology Services India Pvt. Ltd. *Apr 2023 – Apr 2025*

Project: Nadi – Fund Reporting and Management Tool for Private Equity (Daappa)

- Developed and enhanced **Java and Spring Boot** backend services and REST APIs for financial reporting systems and third-party integrations.
- Wrote **Python scripts** for report data processing and automation, supporting fund-reporting and data-validation workflows.
- Adopted **AI-assisted development tools** for code generation, debugging and documentation, improving delivery speed on tight deadlines.
- Built React.js UI enhancements and API integrations; assisted with Dockerized deployments and Jenkins CI/CD on AWS.
- Took part in client discussions, requirement analysis and sprint planning; handled bug fixes, performance optimization and production support.
- **Tech Stack:** Java 17, Spring Boot, Hibernate, Python, React.js, Redux Toolkit, PostgreSQL, REST APIs, Docker, Jenkins, AWS

PROJECTS

DocMind – Chat with Your Documents (RAG) github.com/prasantkumarsahu/docmind

- Built a **retrieval-augmented generation** app: upload PDF/DOCX/MD files and ask questions, with answers grounded in the documents and **cited by page and passage**.
- Async ingestion pipeline with **Spring AI** (PDF/Tika readers, token-based chunking, embeddings) into **pgvector with an HNSW index**; similarity threshold and metadata filters to search selected documents.
- Streams answers over **Server-Sent Events** with per-conversation chat memory, and says so when the answer isn't in the documents instead of hallucinating; runs fully local on **Ollama**.
- **Tech Stack:** Java 21, Spring Boot 4, Spring AI, PostgreSQL + pgvector, Ollama (Llama 3.2, nomic-embed-text), Flyway, React 19, TypeScript, Tailwind CSS, Docker

pg-insight-mcp – Read-only PostgreSQL MCP Server github.com/prasantkumarsahu/pg-insight-mcp

- Built a **Model Context Protocol (MCP)** server in Python that lets LLM clients such as Claude Desktop explore schemas, run analytical SQL and inspect query plans.
- Made it safe by design: **read-only transactions, statement timeouts, row caps**, a SQL guard that blocks writes, DDL and dangerous functions, and a schema allow-list.
- Six structured-output tools plus a schema resource and an analysis prompt; **37 unit and end-to-end tests** against a seeded Postgres database in Docker.
- **Tech Stack:** Python, MCP SDK, psycopg 3, PostgreSQL, pytest, Ruff, Docker

CERTIFICATIONS

- **Professional Scrum Master (PSM I)** – Scrum.org
- **Certified Java & SQL Developer with DSA and System Design** – Geekster.in